

Development of Parallel Stance Check System for Ski Videos Using Pose Estimation Model

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Background and goal of this study

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In ski, Parallel Stance (PS) is pose in which both feet are kept parallel.

This is important for proper ski control.

PS quality has been evaluated by human assessment.

I developed system for objectively checking PS quality from ski videos.

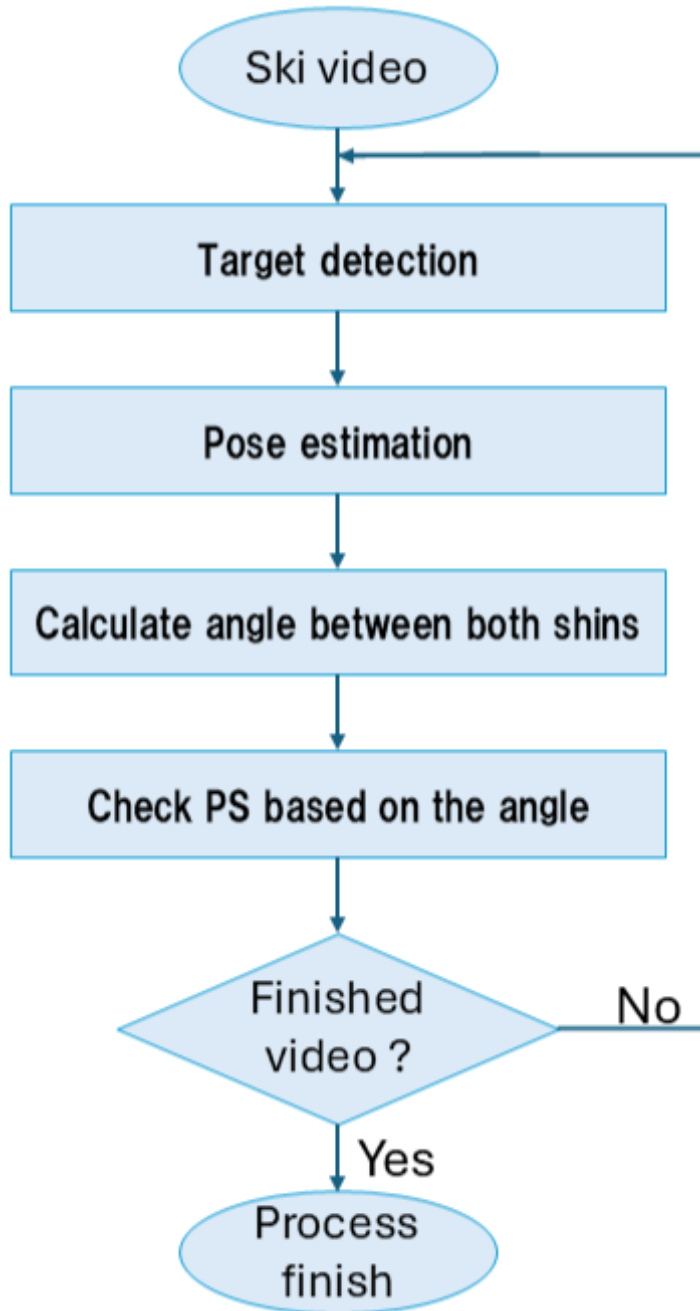


Parallel stance



Not parallel stance

System overview



This is simple flowchart. The key issues are identified.

→ **Tracking may fail due to other skiers or small target.**

→ **Which pose estimation model to use such as 2D and 3D.**

→ **How to check PS based on the angle.**

→ **How to deal with incorrect pose estimation.**

I would like to solve these issues to develop PS check system.

Ski video requirements for reliable pose estimation

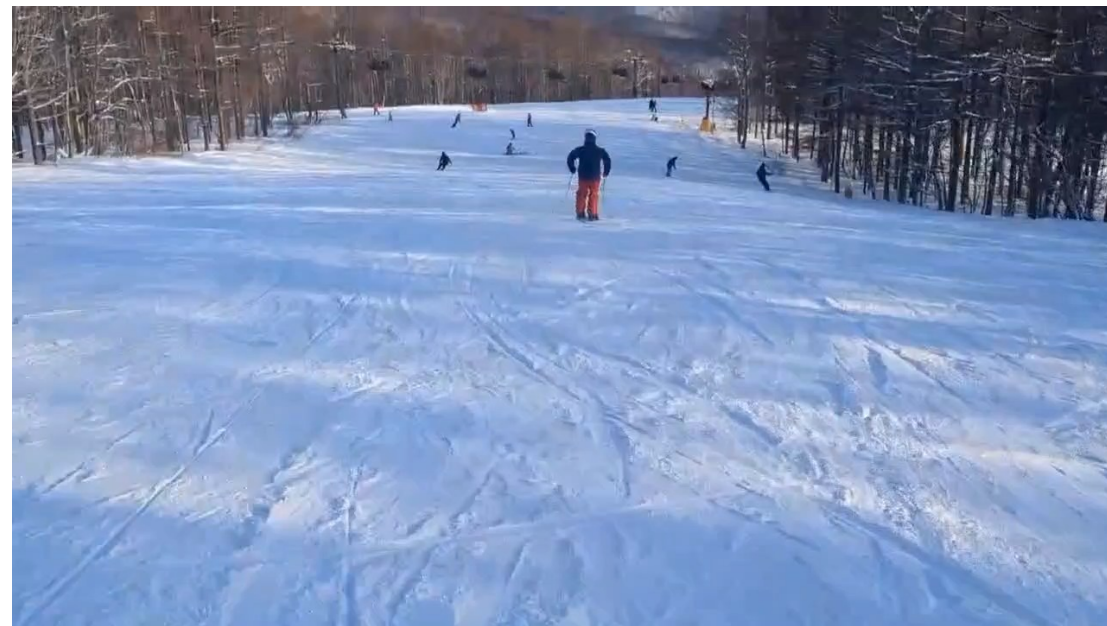
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System requires ski videos to be filmed **from below** or **from behind**.

These viewpoints make pose estimation easier than side, regardless of whether 2D or 3D.



Ski video filmed from below



Ski video filmed from behind

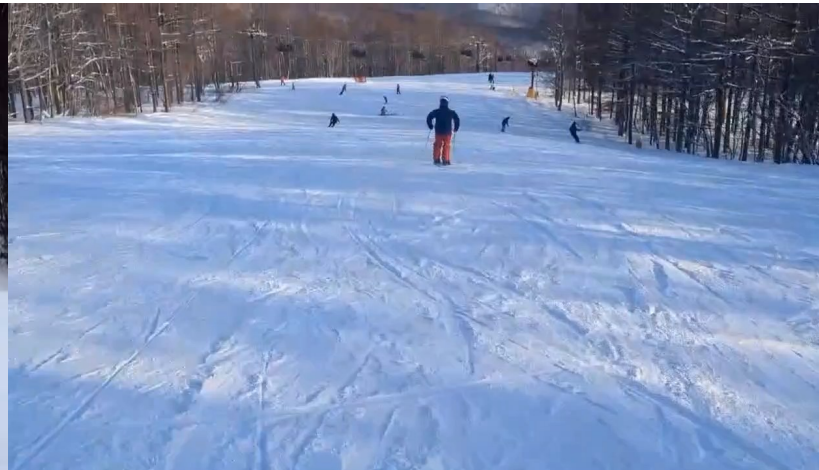
Target detection

Issue 1 : Tracking may fail due to other skiers or small target.

Solutions : Conducting playback direction adjustment based on filming method. Next, I use YOLO (yolo12n.pt) for track target. System design ROI based on bounding box detected in previous frame to track target.



Ski video filmed from below
→ **the largest final frame**
→ **reverse playback**

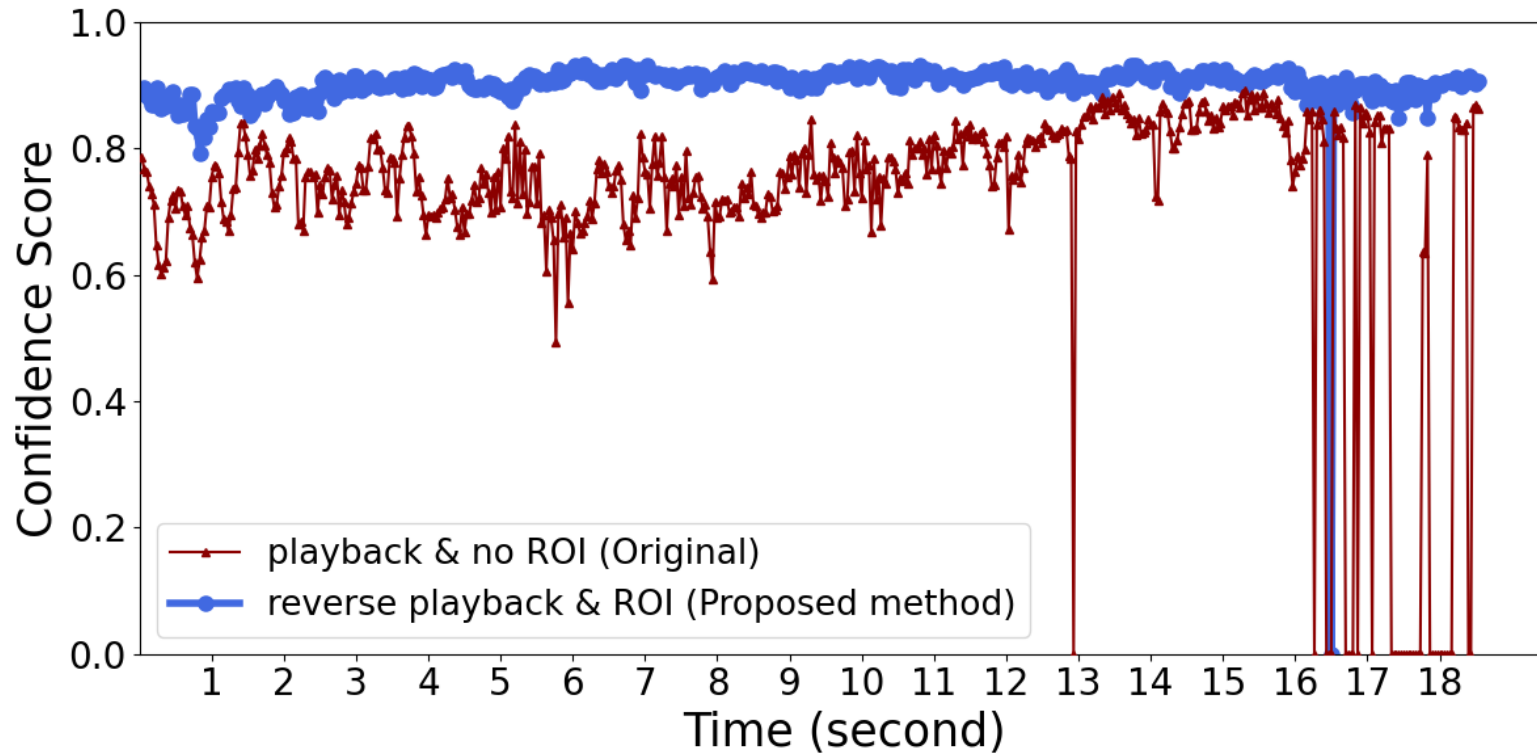


Ski video filmed from behind
→ **playback (no change)**



YOLO is object detection model

Effectiveness of the solutions



Target detection test



Ski video filmed from below

Average confidence
score

Target detection
success rate

Proposed
method
Original

0.904
0.77

99.82 %
93.53 %

Proposed method have
higher average confidence score
and
target detection success rate
than Original.

Pose estimation

Issue 2 : Which pose estimation model to use such as 2D and 3D

Comparison : I compare YOLO (yolo11x-pose.pt), which can estimate 2D pose, and Blaze Pose (pose_landmarker_full.task), which can estimate 3D pose.



Blaze Pose



YOLO

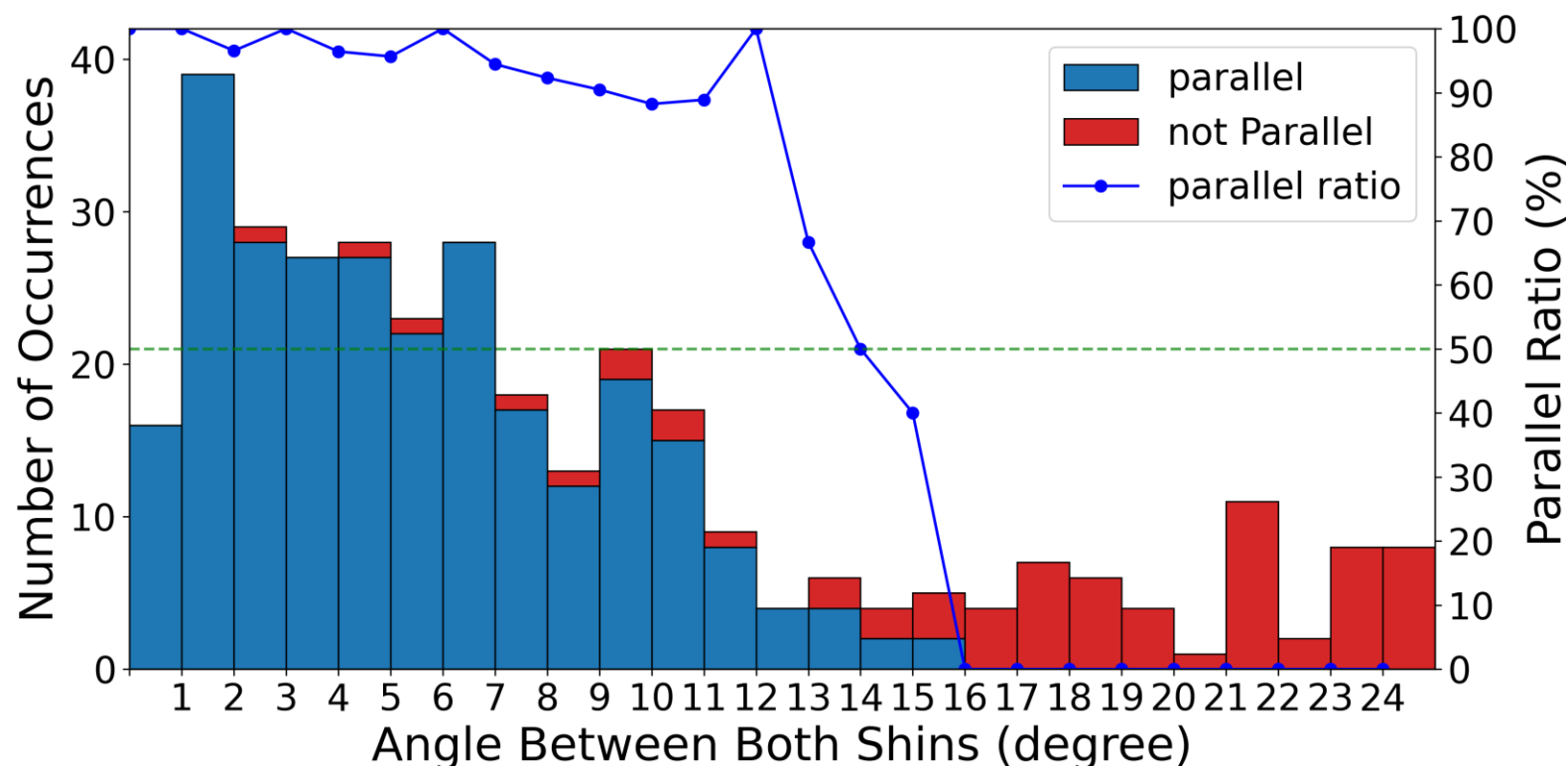
Determination : YOLO can estimate correct pose than Blaze Pose.
I determined to use YOLO in this study.

Check PS based on the angle

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Issue 3 : How to check PS based on the angle.

Solutions : I visualize number of occurrences of parallel and not parallel for each angle using histogram and determine angle thresholds from the histogram.



Less than 13° is almost parallel.

Around 14° to 15° , parallel and not parallel balance is half and half.

More than 16° is perfectly not parallel.

Determination :

The Angle thresholds is:

$\text{angle} < 14^\circ \rightarrow \text{parallel}$

$14^\circ \leq \text{angle} < 16^\circ$
 $\rightarrow \text{controversial form}$

$16^\circ \leq \text{angle}$
 $\rightarrow \text{not parallel}$

Check PS based on the angle

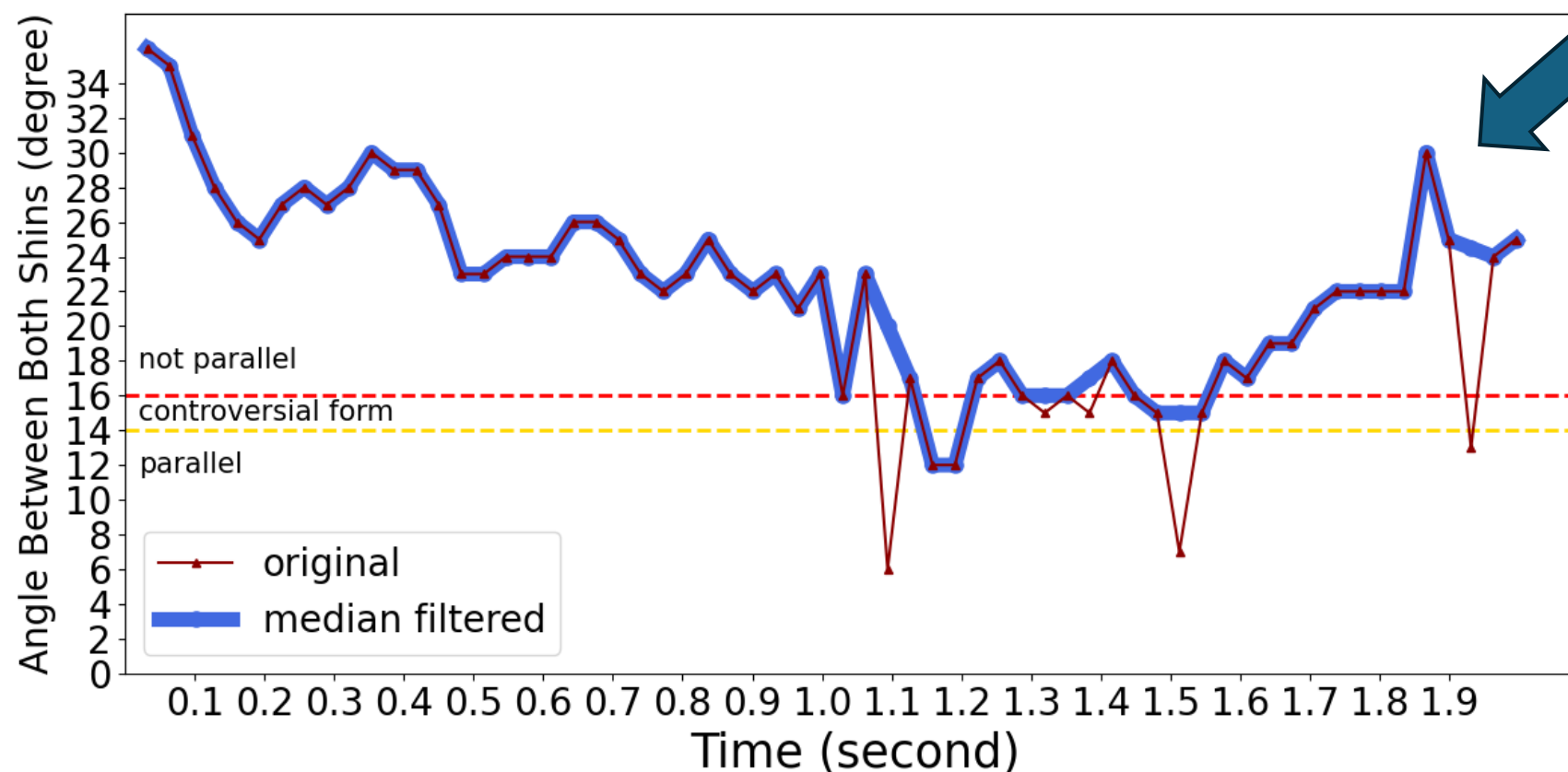
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Issue 4 : Pose estimation models may estimate incorrect pose. In result, system give incorrect judgements, how to deal with it.

Solutions : Cleaning it by median filter.



Ski video of not parallel skier



Effectiveness : After median filtering, incorrect pose estimation of 3 points was cleaned.

Repeatedly incorrect pose estimation of around 1.2 second can't be cleaned.

Developed PS Check System – Front-end (user interface)

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The screenshot shows a web-based user interface for a PS Check System. At the top, there is a text input field labeled 'ファイルの選択' (File Selection) containing the text 'test.mp4'. Below this, a yellow text instruction reads: '動画のトリミングが必要な場合は開始時間と終了時間を指定して下さい。' (If video trimming is necessary, please specify start and end times). The interface includes two time input fields: '開始時間' (Start Time) set to '0' seconds and '終了時間' (End Time) set to '2' seconds, separated by an orange arrow. Below the time fields is a dropdown menu for '撮影方法を選択する' (Select filming method). The dropdown is open, showing two options: '下から撮影' (Filming from below) and '後ろから撮影' (Filming from behind). A blue arrow points to the '下から撮影' option. Below the dropdown, another yellow text instruction reads: 'スキー動画の終了時間時にターゲットがどこにいるか教えて下さい。' (Please tell me where the target is at the end time of the ski video). Below this is a large purple rectangular area representing a video frame, with a smaller red dashed rectangle inside indicating a target region. At the bottom, there is a button labeled 'パラレルスタンスチェック開始' (Start Parallel Stance Check).

User must enter some item to check PS.

1 . Upload a ski video.

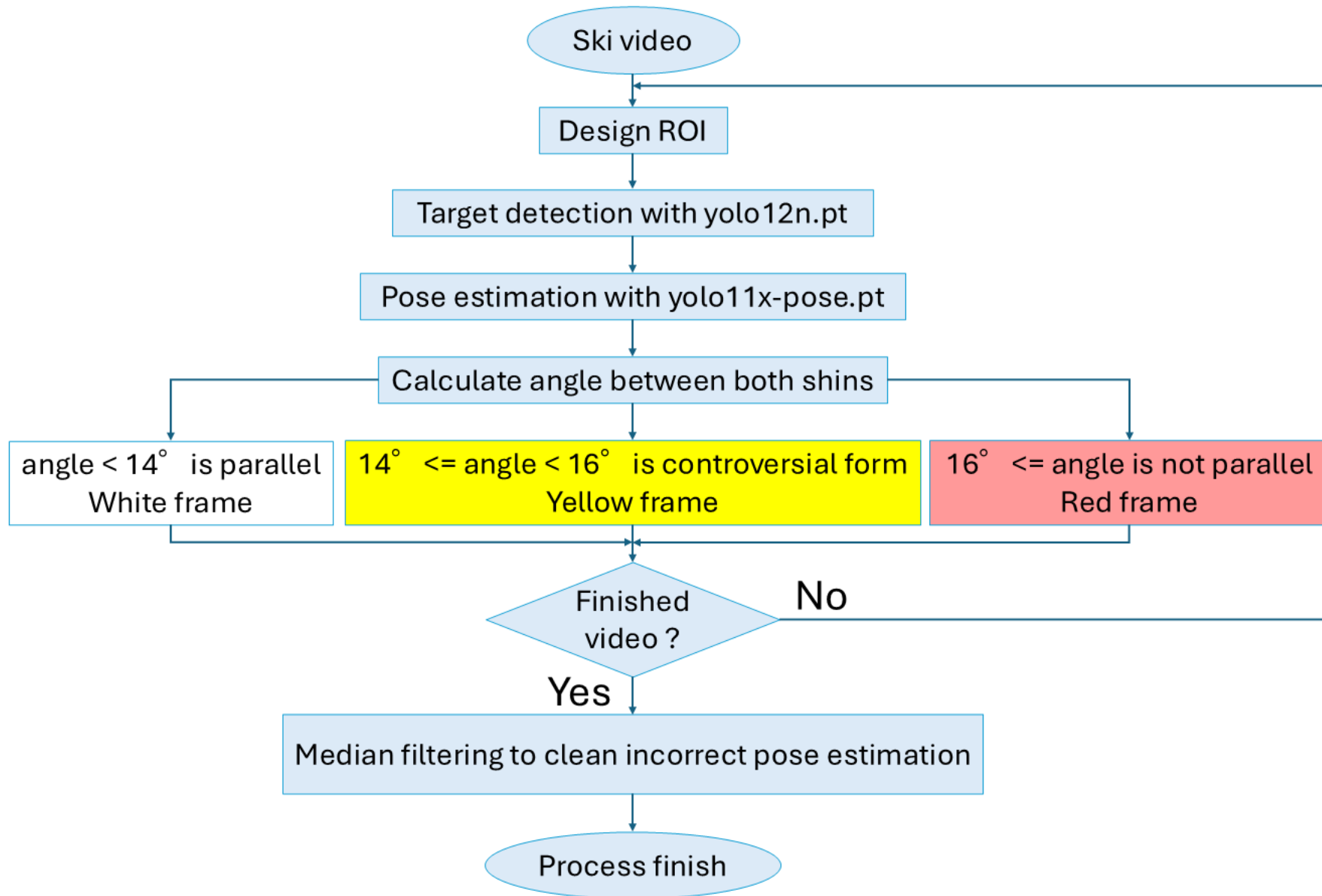
2 . Trim ski video if necessary.

3 . Select filming method.

If user select ski video filmed from below, target is tracked by reverse playback.

4 . Input initial ROI based on filming method. because other skiers may appear.

5. Press start button.



- Back-end program receive ski video and initial ROI from front-end, and process the flowchart.
- To inform user result of PS check, frame is colored based on judgement.

スキー動画のパラレルスタンスチェックシステム

[使い方](#)[開発者](#)

ファイルの選択 **ファイルが選択されていません**

動画のトリミングが必要な場合は開始時間と終了時間を指定して下さい。

開始時間：

秒



終了時間：

秒

撮影方法を選択する ▾

パラレルスタンスチェック開始

What I Did in This Study : I developed system for checking PS quality from ski videos.

What was Achieved : Proposed method can track target by playback direction adjustment and ROI. Pose estimation using YOLO, and perform PS check based on angle between both shins were able to make correct judgment.

Future Challenges : By introducing 3D pose estimation model, relaxing limitations of ski videos and further improving judgement criteria.